IMMUNITY TO MICROBES

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2nd Stage

IBSII /SLG 9

I. INTRODUCTION

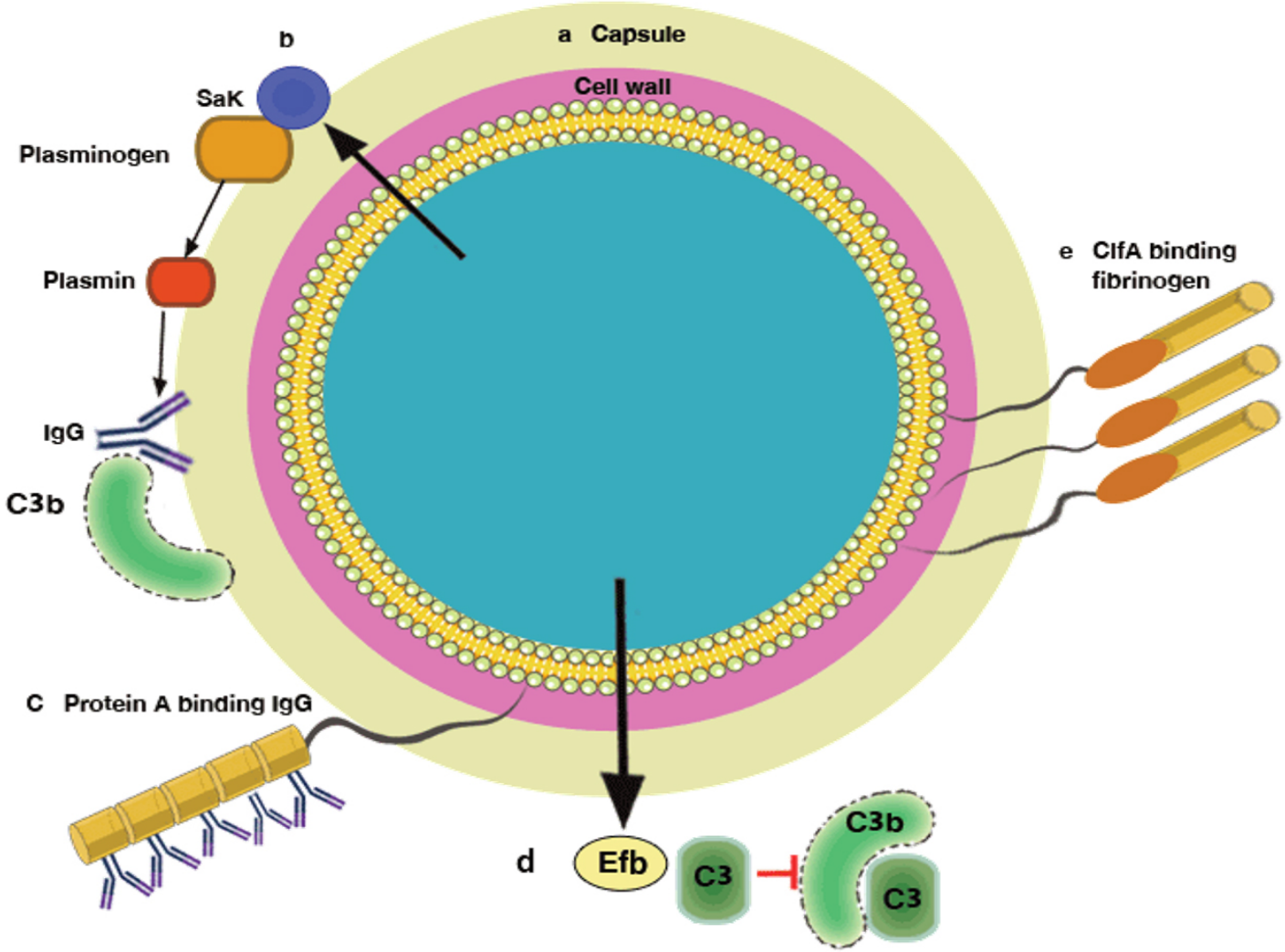
- Pathogens have co-evolved with their hosts, successful pathogens have developed sophisticated mechanisms to
- **avoid recognition,**
- **resist effector mechanisms,**
- or **subvert immune responses**.

I. Immune Evasion by Extracellular Bacteria

- **1. Capsule Formation**
- **Mechanism:** Polysaccharide capsule prevents phagocytosis.
- **Examples:** *Streptococcus pneumoniae*, *Klebsiella pneumoniae*, *Neisseria meningitidis*
- **Effect:**
- Inhibits complement deposition and phagocyte recognition
- Prevents opsonization by antibodies
- Clinical relevance: Severe infections in asplenic and antibody-deficient patients

- **2. Protein A**
- **Mechanism:** Binds Fc region of IgG antibodies.
- **Example:** *Staphylococcus aureus*
- **Effect:**
- Prevents antibody-mediated opsonization
- Blocks Fc receptor-mediated phagocytosis
- Acts as a decoy for antibodies
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Immune System Evasion Mechanisms in *Staphylococcus aureus*: Current Understanding



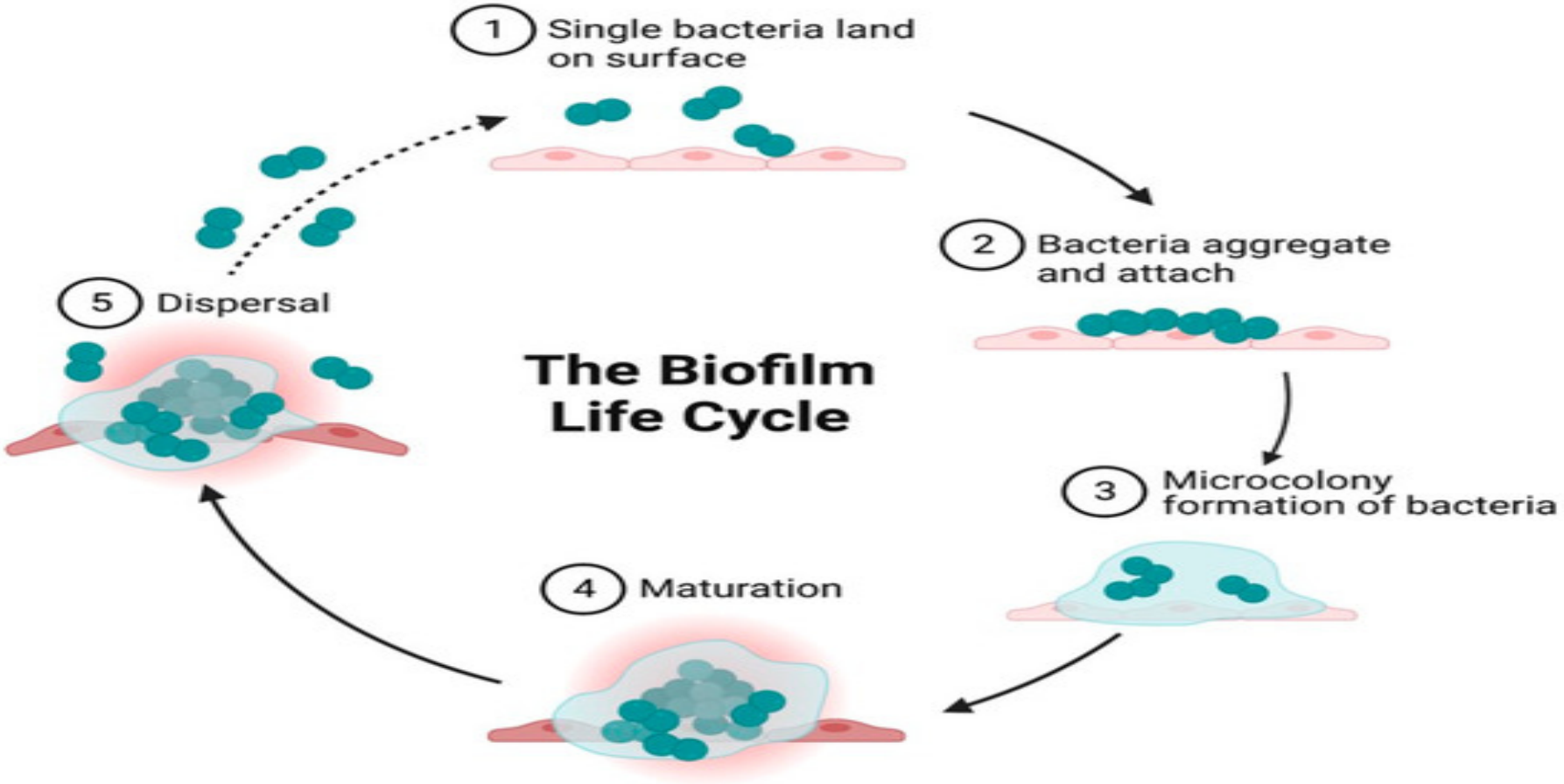
- Mechanisms by which *Staphylococcus aureus* avoids opsonophagocytosis:
- (a) the capsular polysaccharide, which can compromise neutrophil access to bound complement and antibody;
- (b) the extracellular staphylokinase (Sak), which activates cell-bound plasminogen and cleaves IgG and C3b;
- (c) protein A with 5 IgG Fc-binding domains;
- (d) fibrinogen-binding protein (Efb), which binds complement factor C3 and blocks its deposition on the bacterial cell surface. Complement activation beyond C3b attachment is prevented, thereby inhibiting opsonization.
- (e) ClfA, which binds the γ chain of fibrinogen

I. Immune Evasion by Extracellular Bacteria

- **3. Antigenic Variation**
- **Mechanism:** Changes surface antigens to avoid antibody recognition.
- **Examples:** *Neisseria gonorrhoeae* (pili variation),
- **Effect:** Antibodies from previous infection become ineffective.

- **4. Biofilm Formation**
- **Mechanism:** Produces extracellular matrix (polysaccharides, proteins, DNA).
- **Examples:** *Pseudomonas aeruginosa*, *Staphylococcus epidermidis*
- **Effect:**
- Protects against phagocytes
- Resists antibiotics
- Enhances persistence on medical devices (catheters, implants)
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Microbial Biofilm: A Review on Formation, Infection, Antibiotic Resistance, Control Measures, and Innovative Treatment



I. Immune Evasion by Extracellular Bacteria

- **5. Complement Evasion**
- **Mechanism:**
- Recruits host complement regulatory proteins (Factor H)
- Produces complement inhibitors
- **Example:** *Neisseria meningitidis* (Factor H binding protein)
- **Effect:** Prevents complement-mediated lysis and opsonization.

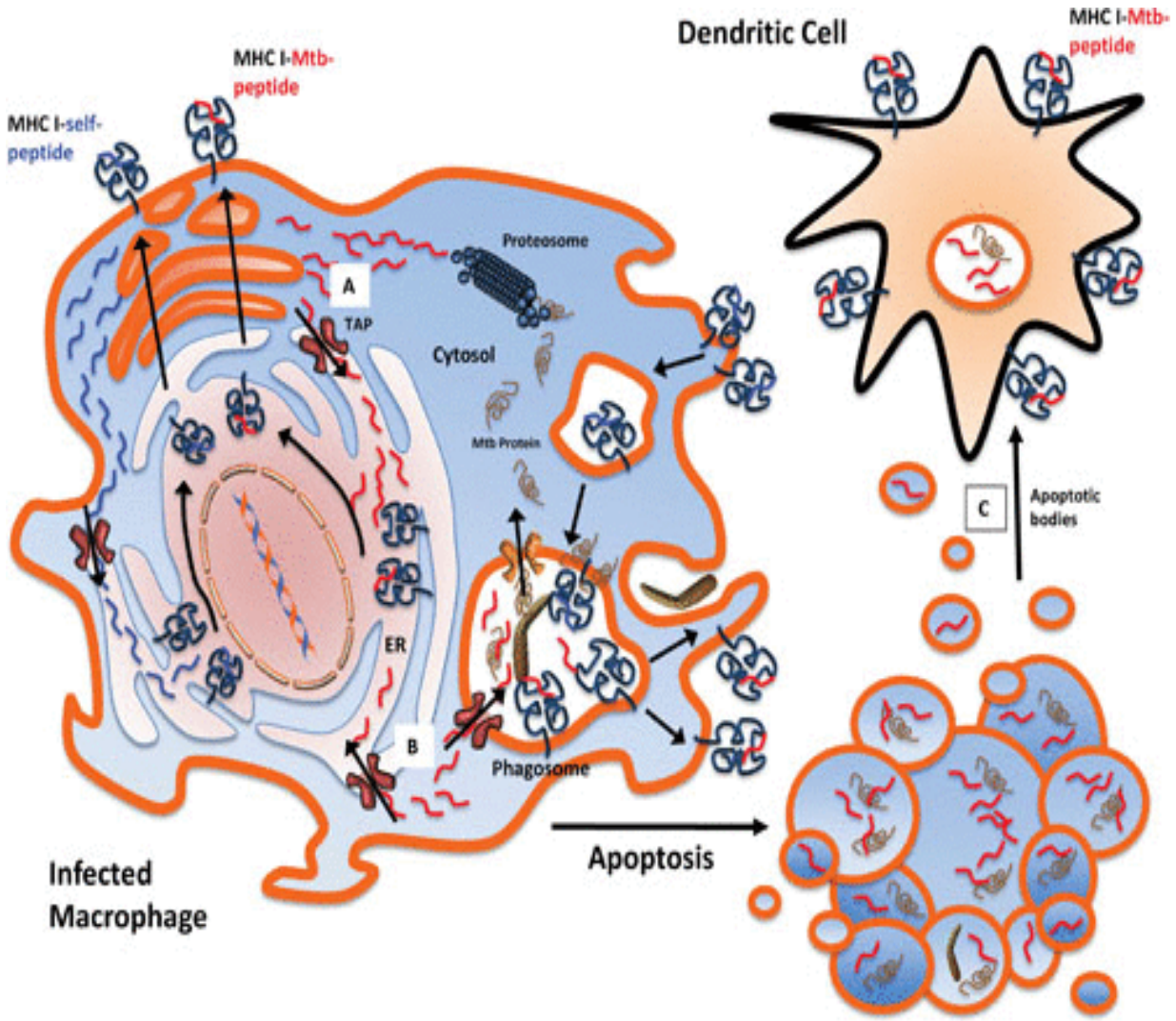
II. Immune Evasion by Intracellular Bacteria

- **1. Inhibition of Phagolysosome Fusion**
- **Mechanism:** Prevents fusion of phagosome with lysosome.
- **Example:** *Mycobacterium tuberculosis*
- **Effect:**
- Bacteria survive in early phagosome
- Avoid exposure to lysosomal enzymes and acidic pH
- Can replicate within the phagosome

- **2. Escape from Phagosome into Cytoplasm**
- **Mechanism:** Lyse phagosomal membrane and escape into cytosol.
- **Example:** *Listeria monocytogenes* (listeriolysin O)
- **Effect:**
- Avoids phagolysosome killing
- Uses host actin for intracellular movement
- Spreads to adjacent cells without exposure to extracellular immunity

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Evasion by *Mycobacterium tuberculosis*



- The peptides from a vacuolar intracellular pathogen such as *M. tuberculosis* can be cross-presented by MHC class I. In this case, an infected cell must first die by apoptosis, and the released apoptotic vesicles carry the mycobacterial antigens into uninfected dendritic cells. these pathways are likely to be actively inhibited or effectively bypassed during *M. tuberculosis* infection

II. Immune Evasion by Intracellular Bacteria

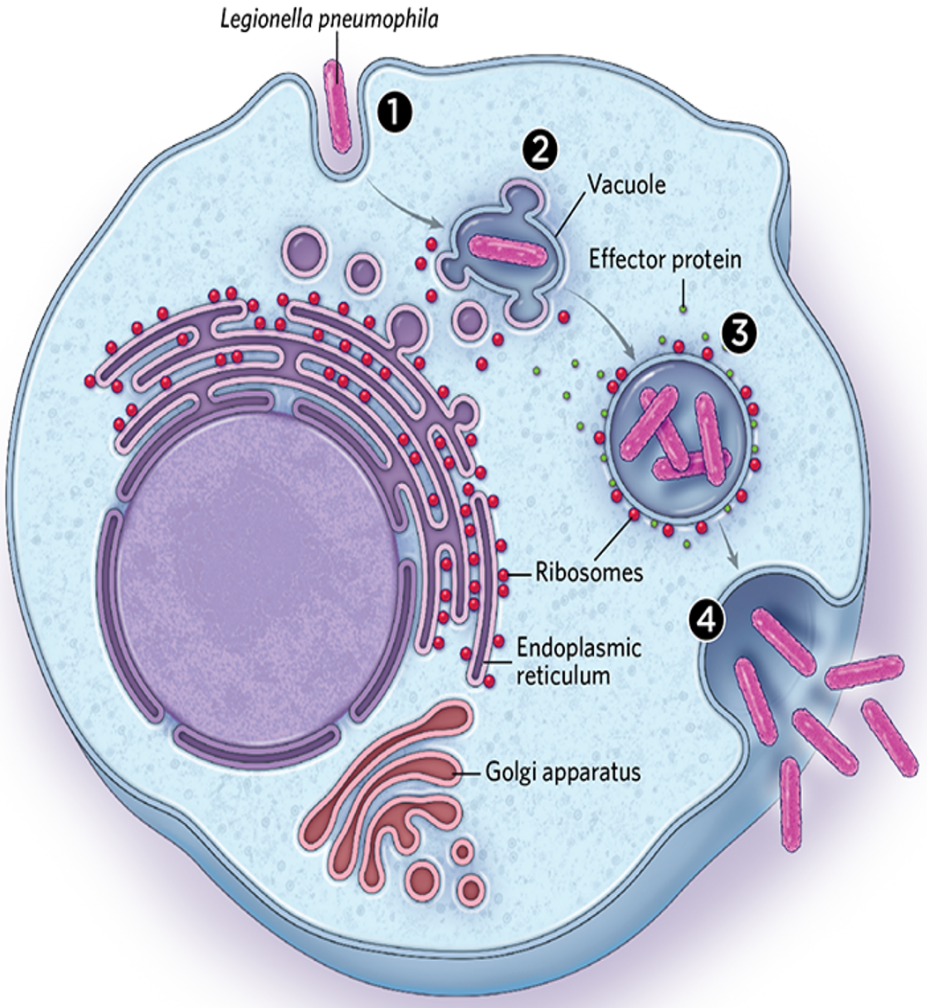
- **3. Inhibition of Apoptosis**
- **Mechanism:** Prevents programmed cell death of infected cells.
- **Examples:** *Chlamydia*, *Mycobacterium*
- **Effect:**
- Prolongs survival of infected cells
- Provides replicative niche
- Avoids immune detection

- **4. Modification of Phagosome Membrane**
- **Mechanism:** Alters phagosome membrane composition.
- **Example:** *Legionella pneumophila* (controls host membranes)
- **Effect:**
- Creates a replicative vacuole
- Recruits host endoplasmic reticulum
- Avoids lysosomal degradation

IMMUNITY TO INTRACELLULAR BACTERIA

- ***Legionella* takes control of host membranes**

  Some intracellular bacteria, such as *Legionella pneumophila*, inhabit membrane-bound compartments inside host cells



III. Immune Evasion by Viruses

- **1. Antigenic Variation**
- **Mechanism:** Changes surface antigens (spike proteins, envelope proteins).
- **Examples:** Influenza (antigenic drift and shift)
- **Effect:**
- Antibodies from previous infection become ineffective
- Requires annual vaccine reformulation
- Allows reinfection
- **Antigenic Drift:** Gradual accumulation of mutations (seasonal influenza)
  Antigenic Shift: Major genetic reassortment (pandemic influenza)

III. Immune Evasion by Viruses

- **2. Production of Decoy Receptors**
- **Mechanism:** Produces soluble receptors that bind cytokines or antibodies.
- **Examples:**
- Poxviruses → soluble cytokine receptors (IL-1β, IFN-γ)
- **Effect:**
- Neutralizes cytokines before they reach immune cells
- Prevents antibody-mediated effector functions
- **3. Inhibition of Apoptosis**
- **Mechanism:** Blocks programmed cell death of infected cells.
- **Examples:**
- Herpesviruses → Bcl-2 homologs
- **Effect:**
- Prolongs survival of infected cells
- Increases viral replication
- Avoids immune clearance

III. Immune Evasion by Viruses

- **4. Latency**
- **Mechanism:** Remains dormant in cells without producing viral proteins.
- **Examples:** Herpes simplex virus (neurons), Varicella-zoster virus (dorsal root ganglia), HIV (CD4+ T cells)
- **Effect:**
- No viral antigens presented
- Invisible to immune system
- Reactivation when immunity wanes

- **5. Formation of Syncytia**
- **Mechanism:** Infected cells fuse with neighboring cells.
- **Examples:** HIV, RSV, measles
- **Effect:**
- Spreads infection without exposure to extracellular antibodies
- Avoids neutralization

IV. Immune Evasion by Parasites

- **1. Antigenic Variation**
- **Mechanism:** Changes surface antigens to avoid antibody recognition.
- **Examples:**
- *Trypanosoma brucei* → Variant Surface Glycoprotein (VSG) switching
- **Effect:**
- Antibodies from previous wave ineffective
- Chronic infection with repeated waves
- Immune system constantly chasing new variants

- **2. Intracellular Residence**
- **Mechanism:** Lives inside host cells.
- **Examples:**
- *Leishmania* → macrophages
- **Effect:**
- Protected from antibodies and complement
- Avoids extracellular immune mechanisms
- Requires cell-mediated immunity for clearance

IV. Immune Evasion by Parasites

- **3. Complement Evasion**
- **Mechanism:**
- Recruits host complement regulatory proteins
- Produces complement inhibitors
- **Examples:**
- *Schistosoma* → produces complement inhibitors
- **Effect:** Prevents complement-mediated lysis.

- **4. Formation of Cysts**
- **Mechanism:** Forms dormant cysts resistant to immune attack.
- **Examples:** *Entamoeba histolytica*, *Giardia lamblia*, *Toxoplasma gondii*
- **Effect:**
- Survives in environment
- Resists immune clearance
- Reactivation when immunity wanes

V. Immune Evasion by Fungi

- **1. Capsule Formation**
- **Mechanism:** Polysaccharide capsule prevents phagocytosis.
- **Example:** *Cryptococcus neoformans*
- **Effect:**
- Inhibits phagocytosis
- Reduces complement deposition
- Causes meningitis in immunocompromised

- **2. Biofilm Formation**
- **Mechanism:** Produces extracellular matrix on surfaces.
- **Example:** *Candida albicans* on catheters and medical devices
- **Effect:**
- Resists phagocytes
- Resists antifungal drugs
- Persistent infection

V. Immune Evasion by Fungi

- **3. Intracellular Survival**
- **Mechanism:** Survives inside macrophages.
- **Examples:** *Histoplasma capsulatum*, *Cryptococcus*
- **Effect:**
- Protected from antibodies and complement
- Avoids extracellular immune mechanisms
- Requires Th1 immunity for clearance

- **4. Dimorphism**
- **Mechanism:** Switches between yeast and hyphal forms.
- **Examples:** *Candida albicans*, *Histoplasma capsulatum*
- **Effect:**
- Yeast form → dissemination
- Hyphal form → tissue invasion
- Immune recognition varies by form

VI. Immune Evasion by Prions

- **1. Recognition as "Self"**
- **Mechanism:** PrP^Sc is an abnormal conformation of normal host protein (PrP^C).
- **Effect:**
- No antibodies generated
- No T cell responses
- No inflammation
- No immune response

- **2. Lack of Antigen Presentation**

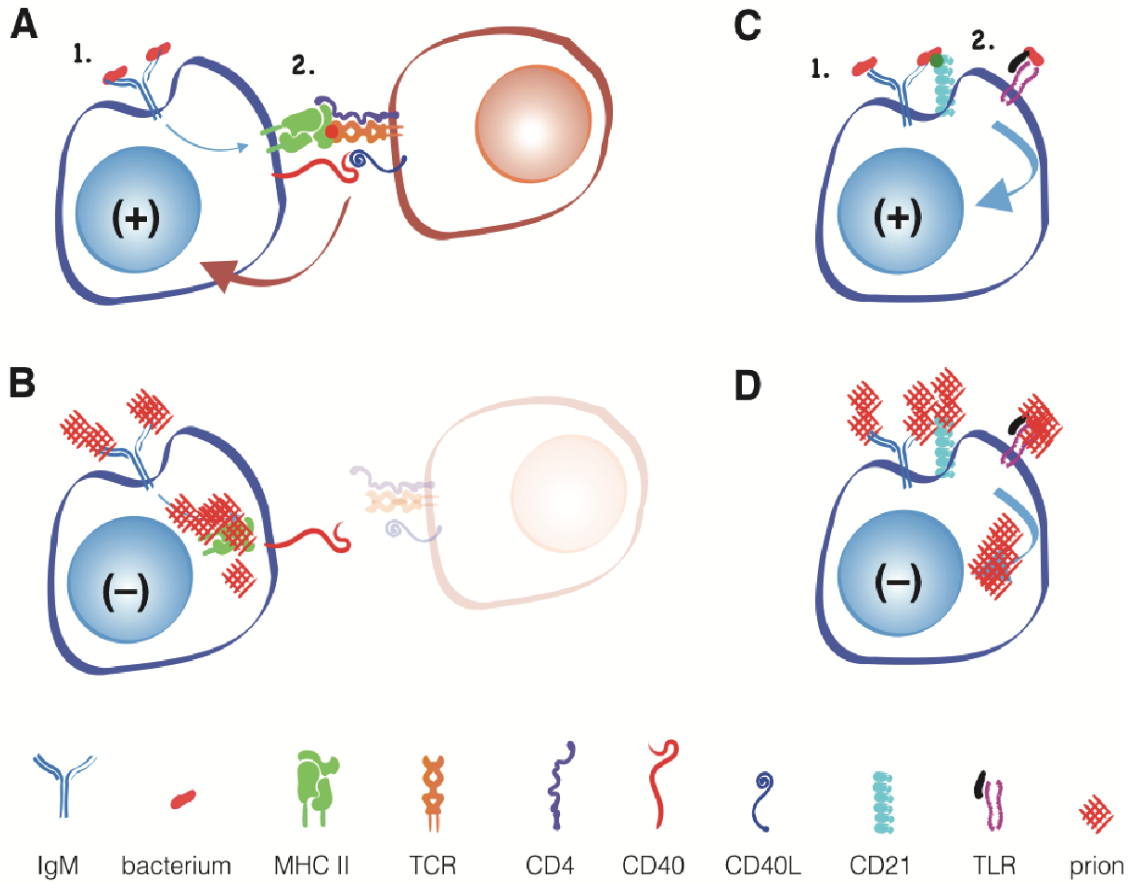
- **3. Failure to Activate B Cells**

VI. Immune Evasion by Prions

- **4. Absence of Inflammation**
- **Mechanism:** No innate immune activation.
- **Effect:**
- No inflammatory response
- No recruitment of immune cells
- Silent neurodegeneration

- **5. Resistance to Degradation**
- **Mechanism:** PrP^Sc is highly resistant to proteases and degradation.
- **Effect:**
- Accumulates in brain
- Causes neurodegeneration
- Cannot be cleared by immune mechanisms

- Prions fail to activate B cells with or without T cell help. A simplified model of T cell-dependent and independent B cell activation. Binding of antigen to IgM on the surface of B cells provides the first activation signal.
- (A) T cells that recognize antigen processed by the B cell and presented in MHC II molecules provide the second signal, typically through its CD40 ligand binding CD40 on the B cell during infections with classic pathogens.
- (B) Prion-specific T cells are absent during prion infection. Other innate immune receptors, like CD21 or TLRs, can provide secondary signals without T cells in response to
- (C) bacteria, but not to
- (D) prions.



Immune Evasion

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Pathogen Type	Key Evasion Strategies
Extracellular bacteria	Capsules, Protein A, antigenic variation, biofilms, complement evasion
Intracellular bacteria	Inhibit phagolysosome fusion, escape to cytoplasm, resist oxidative killing, downregulate MHC, inhibit apoptosis, inhibit autophagy
Viruses	Antigenic variation, MHC-I downregulation, decoy receptors, latency, immune-privileged sites, inhibit apoptosis, interfere with IFN, molecular mimicry, syncytia formation
Parasites	Antigenic variation, intracellular residence, immunosuppression, molecular mimicry, complement evasion, cysts, anatomical sequestration
Fungi	Capsules, biofilms, intracellular survival, dimorphism, melanin, inhibit phagolysosome fusion, modulate cytokines, proteases, siderophores
Prions	"Self" protein; no immune response; no antigen presentation; no inflammation; failure to activate B cells; resistant to degradation

Immune Responses to Major Pathogens

Pathogen	Innate Immunity	Adaptive Immunity	Key Cytokines
Extracellular bacteria	Complement, phagocytosis, inflammation	Antibodies (opsonization, neutralization), Th17	IL-17, TNF-α
Intracellular bacteria	NK cells, macrophages	Th1 (IFN-γ), CD8+ CTLs, granuloma	IFN-γ, TNF-α
Viruses	Type I IFNs, NK cells	CD8+ CTLs, antibodies, Th1	IFN-α/β, IFN-γ, IL-2
Parasites (helminths)	Eosinophils, mast cells	Th2, IgE, IgA	IL-4, IL-5, IL-13
Parasites (protozoa)	Macrophages, NK cells	Th1, CD8+ CTLs	IFN-γ
Fungi	Neutrophils, macrophages	Th1, Th17	IFN-γ, IL-17
Prions	None effective	None effective	None

References

- Abbas, A. K., Lichtman, A. H., & Pillai, S. (2022). *Cellular and Molecular Immunology* (10th ed.). Elsevier. — Chapter 16: Immunity to Microbes.
- Murray, P. R., Rosenthal, K. S., & Pfaller, M. A. (2021). *Medical Microbiology* (9th ed.). Elsevier. — Chapter 10: Immune Responses to Infectious Agents.
- Playfair, J. H. L., & Bancroft, G. J. (2013). *Infection & Immunity* (4th ed.). Oxford University Press.
- Kuby, J., et al. *Kuby Immunology* — Part IV: Immunity to Viruses and Bacteria; Immunity to Fungi and Parasites.
- *Mims' Medical Microbiology and Immunology* (7th ed.). Elsevier.
- Finlay, B. B., & McFadden, G. (2006). Anti-immunology: evasion of the host immune system by bacterial and viral pathogens. *Cell*, 124(4), 767–782.
- Alcami, A., & Koszinowski, U. H. (2000). Viral mechanisms of immune evasion. *Trends in Microbiology*, 8(9), 410–418.